MILITARY SPECIFICATION

LADDERS, VEHICLE BOARDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of vehicle boarding ladders.

1.2 Classification. Ladders shall be of the following types, as specified (see 6.2):

Type I - Ladder, vehicle boarding, 57.50 inches overall length.
Type II - Ladder, vehicle boarding, 65 inches overall length.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet.
QQ-S-634	- Steel, Bar, Carbon, Cold Finished, (Standard Quality).
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
QQ-S-741	- Steel, Carbon: Structural Shapes; Plates; and Bars.
RR-G-1602	- Grating, Metal, Other Than Bar Type (Floor, Except for Naval Vessels).

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MS20613	- Rivet, Solid-Universal, Head, Steel, Carbon, and Steel, Corrosion-Resistant.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY (AWS)

A 2.0 Welding Symbols.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The ladders shall be as specified herein. Type I ladders shall be as shown in Figures 1, 3, and 5. Type II ladders shall be as shown in Figures 2, 4, and 5.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Sheet steel. Sheet steel shall conform to QQ-S-698, cold rolled, commercial quality, Composition 1008 to 1020, .075 inch thick.

3.2.2 Angles. Angles shall be fabricated from steel conforming to QQ-S-741.

3.2.3 Steel bars. Steel bars shall conform to QQ-S-634, cold finished condition, Composition 1010 to 1020. Rods shall be furnished in round form, and plates shall be furnished in flat form.

3.2.4 Aluminum alloy gratings. Aluminum alloy gratings shall conform to RR-G-1602, Type IV, Alloy 5052, and shall be 7 inches wide with three diamond-shaped openings across the panel.

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3.2.5 Aluminum alloy sheet. Aluminum alloy sheet shall conform to QQ-A-250/8, Temper 0. General requirements for this alloy sheet are found in QQ-A-250.

3.3 Rivets. Rivets shall conform to MS20613-12C14.

3.4 Identification marking. The ladders shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The portions of the ladder normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Color shall be as specified (see 6.2).

3.6 Workmanship.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by commercially approved methods that will not injure the material. Corners shall be square. Flame cutting, using tips suitable for the thickness of steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs and sharp edges.

3.6.2 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas (argon or helium) shielded-arc method. Welds may be ground, filed, buffed, or chipped but not hammered and shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter. For interpretation of welding symbols, see AWS A 2.0.

3.6.3 Steel welding. The surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, mill scale, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the ladder

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to conform to the examination and test requirements specified in Section 4. Unless otherwise specified (see 6.2), the welding process used in fabrication of the ladder shall be at the option of the supplier.

3.6.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL

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shall be 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Inspection procedure.

4.4.1 Examination. The ladders shall be examined as specified herein for the following defects:

Major

101. Ladders not as shown in Figures 1 through 5 as applicable.
102. Components missing or not as specified.
103. Materials not as specified.
104. Workmanship not as specified.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Containers not as specified.
106. Number of ladders packed together in one container not as specified for Level A or B.
107. Cushioning not as specified for Level A or B.
108. Marking illegible, incorrect, or incomplete.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A. The ladders shall be packed in wood boxes conforming to PPP-B-601, Overseas Type, Grade B or PPP-B-621, Class 2. The number of each type ladder to be packed in one box shall be as specified (see 6.2). Cushioning shall be placed between the points of contact of the ladders to prevent abrasion.

5.1.2 Level B. The ladders shall be packed as specified in 5.1.1 for Level A except the boxes shall be Domestic Type.

5.1.3 Level C. The ladders shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I ladder. The Type I ladder is intended to provide access to the rear entrance of a Size 17 truck van. NSN 2540-00-355-6376 which is used in Equipment Sets 3610-97-CL-E11, 3610-97-CL-E12, 3610-97-CL-E13, 3610-97-CL-E14, 6675-97-CL-E35, 6675-97-CL-E38, and 6675-97-CL-E45 applies thereto. This specification is intended to replace CE PD-288, dated 13 September 1955.

6.1.2 Type II ladder. The Type II ladder is intended to provide access to the side entrance of a Size 17 truck van. NSN 2540-00-346-2927 which is used in Equipment Sets 3610-97-CL-E11, 3610-97-CL-E12, 3610-97-CL-E13, 3610-97-CL-E14, 6675-97-CL-E35, and 6675-97-CL-E38, applies thereto. This specification is intended to replace CE PD-288, dated 13 September 1955.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Color required (see 3.5).
(d) Welding process to be used when other than at the option of the supplier (see 3.6.3).
(e) Level of packing required (see 5.1).
(f) Number of ladders of each type to be packed together (see 5.1.1).

Custodian:

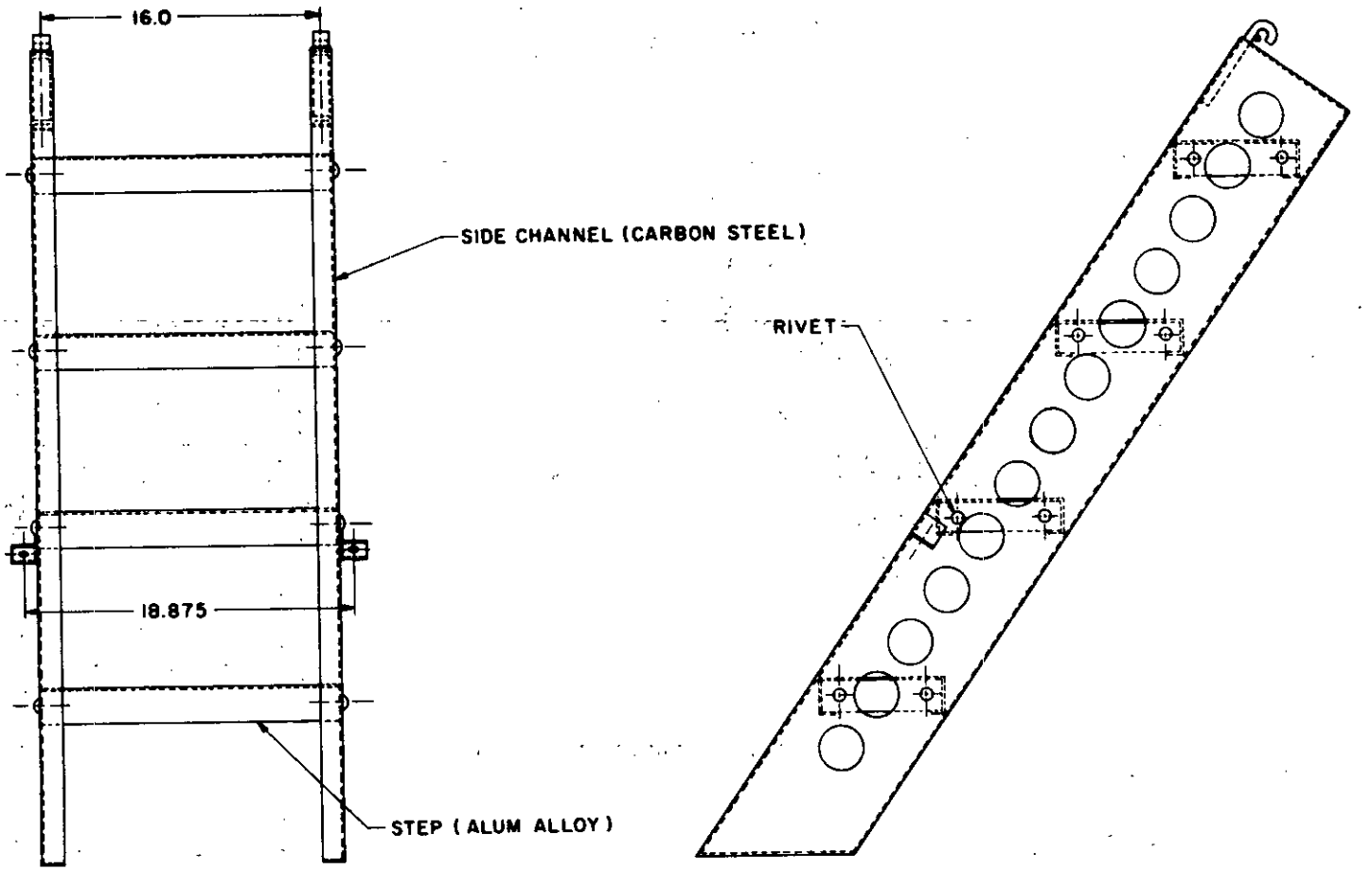
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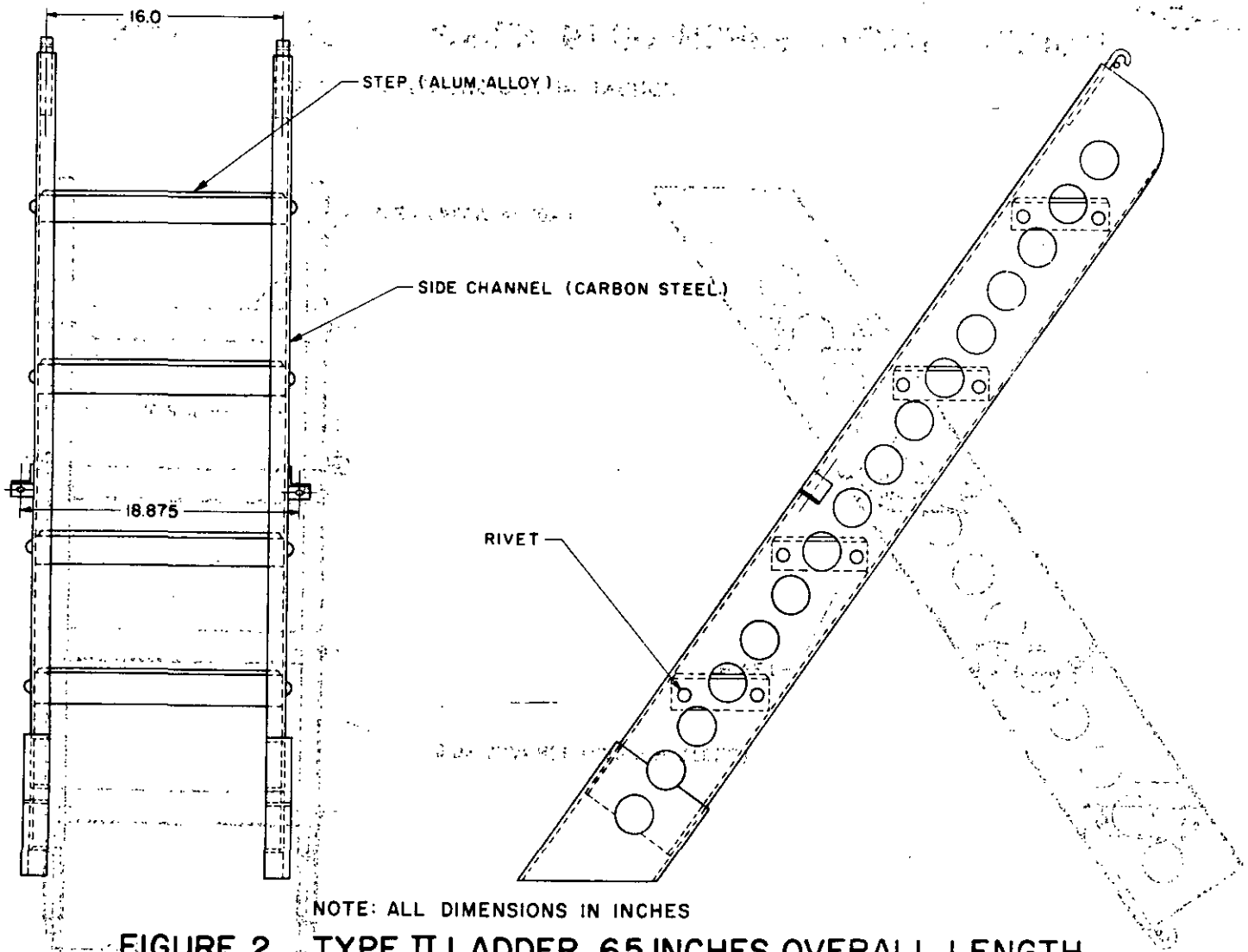


NOTE: ALL DIMENSIONS IN INCHES

FIGURE 1. TYPE I LADDER, 57.50 INCHES OVERALL LENGTH

X-2724

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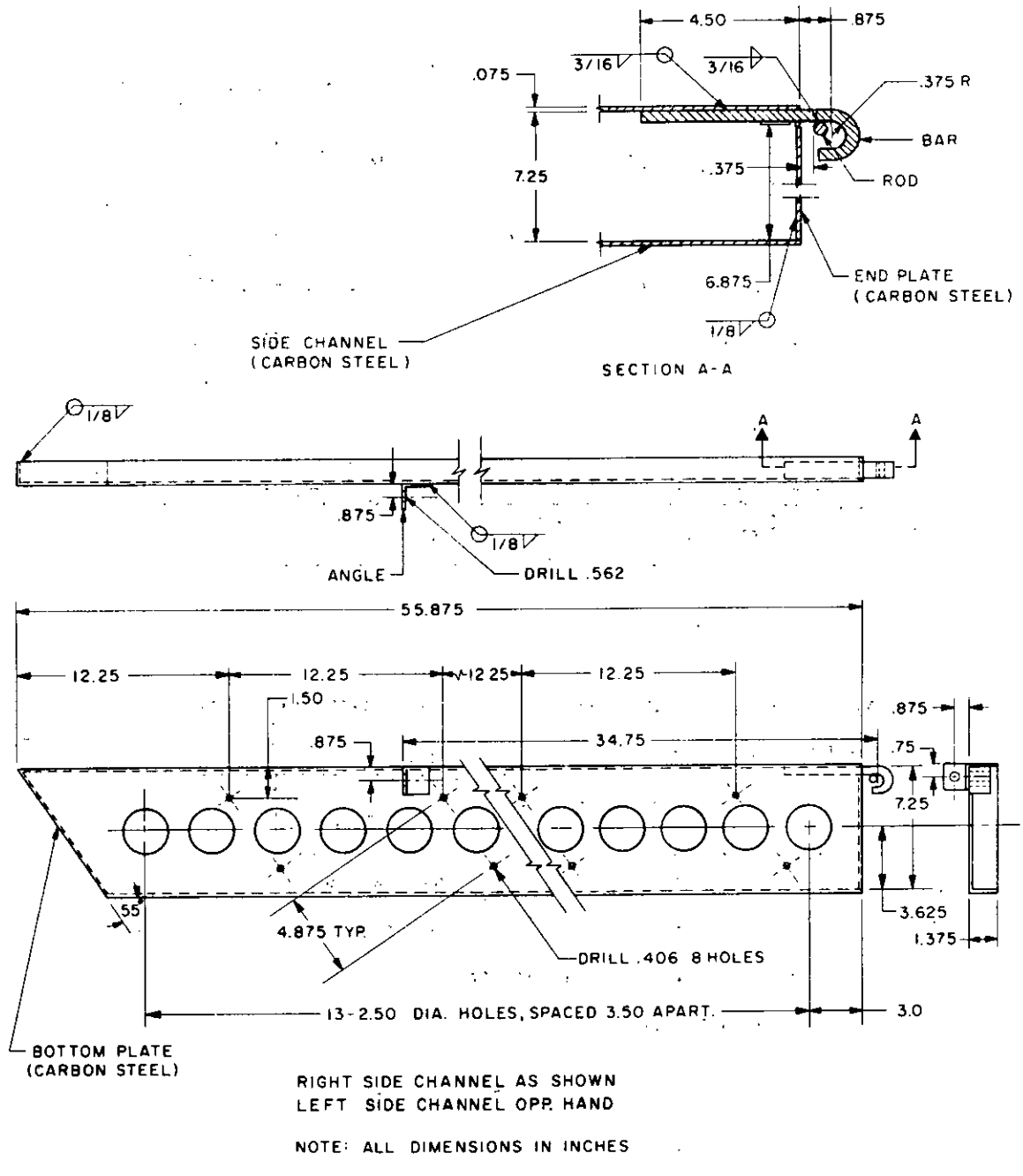


NOTE: ALL DIMENSIONS IN INCHES

FIGURE 2. TYPE II LADDER, 65 INCHES OVERALL LENGTH

X-2725

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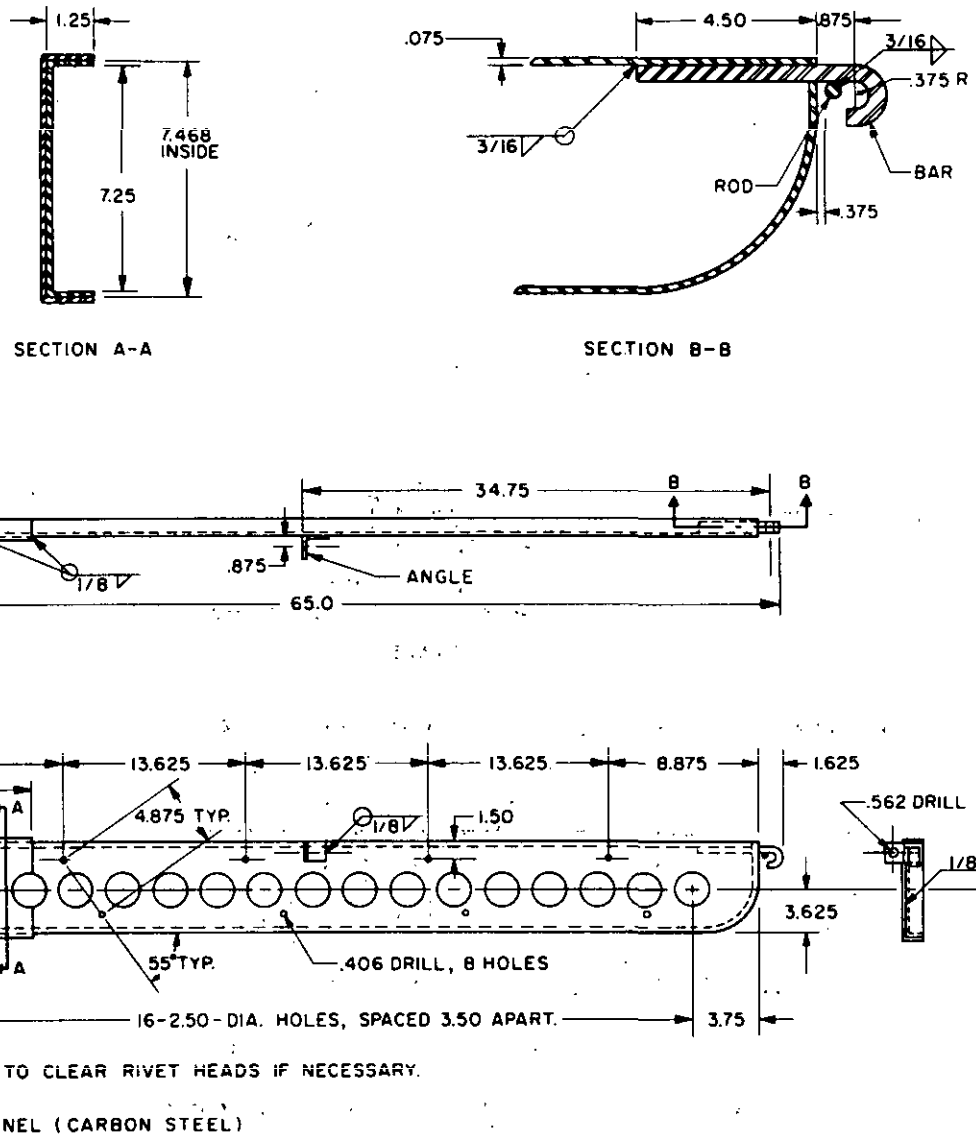
RIGHT SIDE CHANNEL AS SHOWN
LEFT SIDE CHANNEL OPP. HAND

NOTE: ALL DIMENSIONS IN INCHES

FIGURE 3. TYPE I LADDER, SIDE CHANNEL

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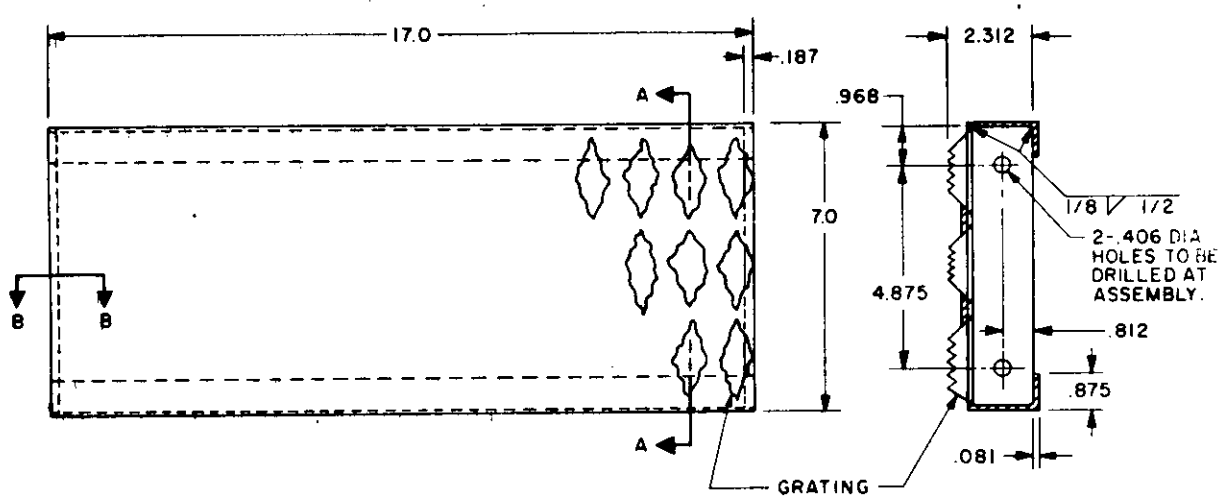
RIGHT SIDE CHANNEL- AS SHOWN
LEFT SIDE CHANNEL- OPP. HAND

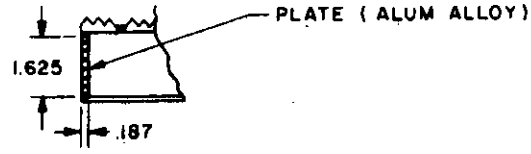
NOTE: ALL DIMENSIONS IN INCHES

FIGURE 4. TYPE II LADDER, SIDE CHANNEL

X-2727

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NOTE: ALL DIMENSIONS IN INCHES.

FIGURE 5. TYPE I AND TYPE II LADDERS, STEP

X-2728

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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